Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination
October -2023

Max. Marks: 30 | **Duration: 01 hour**
Class: B.Tech | **Semester: V**
Course Code: CS-304 | **Branch: Computer Engineering**
Name of the Course: Distributed Computing

Instructions:
(1) All Questions are Compulsory.
(2) Draw neat diagrams.
(3) Assume suitable data if necessary.

Q. No.	QUESTION	Max. Marks	CO-BL
Q.1	What is the scalability in distributed systems? Which are the different dimensions to measure the scalability of a distributed system? What are the scalability problems and solutions/techniques? Explain with example.	06	CS304.1-2
Q.2	A multicomputer with 256 CPUs is organized as 16 × 16 grid. What is the worst case delay (in hops) that a message might have to take?	04	CS304.1-5
Q.3	In this problem, you are to compare reading a file using a single-threaded file server and a multithreaded server. It takes 15 msec to get a request for work, dispatch it, and do the rest of the necessary processing, assuming the data needed are in a cache in main memory. If a disk operation is required, as is the case one-third of the time, an additional 75 msec is required, during which time the thread sleeps. How many requests/sec can the server handle: a. if it is single-threaded? b. If it is multithreaded?	05	CS304.3-4
Q.4	What are the different ways to organize servers? What are the different approaches to get the endpoint in the design issues of identifying the endpoint?	05	CS304.3-2
Q.5	What are the advantages and drawbacks of using LWP's (Lightweight Process) in combination with a user-level thread package?	05	CS304.3-2
Q.6	How would a receiving process in a nonblocking receive() primitive know when a message arrives in the message buffer? What are the methods used for this purpose? **OR** Give two examples of application for which each of the following types of multicast communication is most suitable. a. The 0-reliable b. The 1-reliable c. The m-out-of-n reliable (1<m<n) d. All-reliable	05	CS304.2-2